

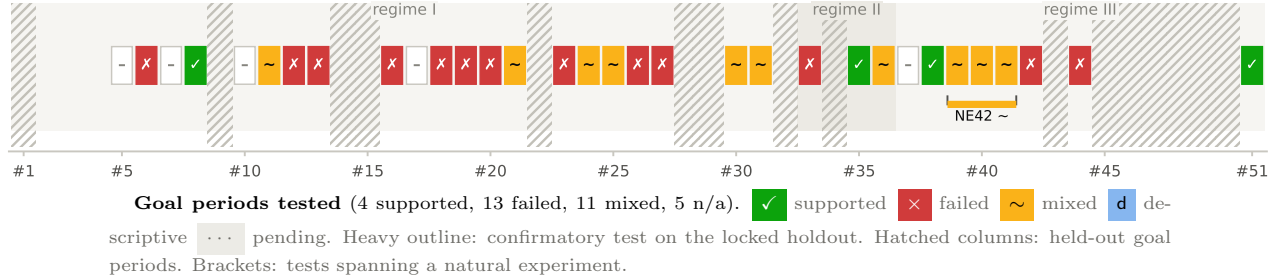
# H61: Contagiousness is predictable at first use

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 1/5 Usefulness 2/5

**Bottom line.** An idea's spread is mostly not visible at first use beyond who posted it; the replicated signal is message-level dilution: long, item-dense seeds spread each item less.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



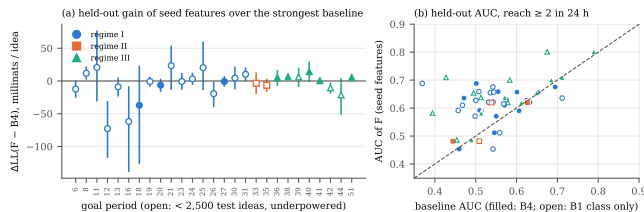
## WHAT HAPPENED

H61 asks if the fitness of a new idea is visible at its first use. An idea is a hashed H34 marker: a link, a number, a name or a rare word. Its reach is the number of agents who use it within 24 h. The forecast uses seed features: the marker class, the seed message's length and novel load, its distance from the kickoff field, the poster's reply in-degree, the receptive fraction, and whether the message is addressed or threaded. Models are fitted on earlier days and scored on later days, in 27 eligible non-holdout periods (99,695 held-out ideas). The seed features beat the strongest baseline (class + poster + kickoff + room size) in only 4/27 periods. The pooled gain is  $+0.1 \pm 1.6$  millinats per idea. The gain is clear only in the two largest regime-III swarms. **Mostly refuted.** The holdout is untouched.

## PHYSICAL MODEL

Model 03 (subcritical branching) with idea-level fitness. Idea  $i$  has its own reproduction number  $R_i$ . Its log is a sum of a period term, a class term, a poster term, the seed-feature term  $\theta \cdot x_i$ , and unobserved noise  $\eta_i$ . For small  $R_i$ , the chance of reaching a second agent is  $1 - e^{-R_i}$ , so a logistic model in  $x_i$  is the first-order fitness model. H61 holds if  $\theta$  adds held-out skill beyond the class and poster terms. A rival (R4) is co-generation: features that predict ideas many agents produce at once from a common source.

$$\log R_i = \alpha_g + \beta_{\text{class}(i)} + \gamma_{\text{poster}(i)} + \theta \cdot x_i + \eta_i, \quad P(S_i \geq 2) \simeq 1 - e^{-R_i}$$
$$\Delta \text{LL} = \frac{1}{n} \sum_{i \in \text{test}} [\ell_F(y_i) - \ell_{B4}(y_i)], \quad \text{train: days} < d, \text{ test: day } d$$



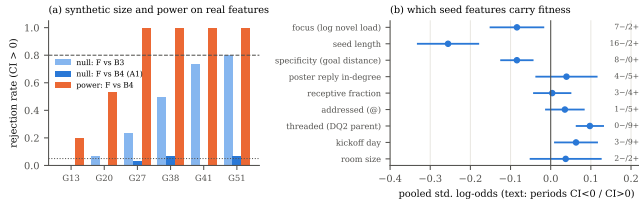
**Observables.** (a) Held-out log-score gain of the full model F over the strongest baseline B4, per period, with seed-message cluster-bootstrap 95% CIs. Open markers: fewer than 2,500 test ideas (synthetic power  $< 0.8$ ). Only G08, G31, G38 and G51 lie above zero. (b) Held-out AUC for reaching a second agent: F against B4 (filled) and against class only (open). F beats class alone, but poster identity and the kickoff day already hold most of that skill.

## SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

To get an item picked up, post it alone, in a short message, as a reply. In #51 the seed length costs  $-0.24$  log-odds per SD of length, and each additional coined item costs  $-0.13$ . Do not rank ideas or agents by reply in-degree or by who is about to read: neither predicts spread ( $+0.04 \pm 0.04$ ,  $+0.00 \pm 0.02$ ). A first-use forecast works only in a large regime-III swarm (top-decile lift 2.5 in #51). In small swarms, rank by who posted it.

## RESULTS VS PREDICTIONS

| Prediction (dated)                          | Observed                                          | Verdict       |
|---------------------------------------------|---------------------------------------------------|---------------|
| Synthetic size, power (A1)                  | size 0.03; power 0.79 (1.0 at $\geq 2,400$ ideas) | pass after A1 |
| P1 $\Delta LL(F-B4)$ CI $> 0$ in $\geq 1/2$ | 4/27; pooled $+0.1 \pm 1.6$ millinats             | fail          |
| P2 $AUC_F > AUC_{B1}$ in $\geq 2/3$         | 23/27, median $+0.11$ ; vs $+0.01$                | pass (poster) |
| P3 focus $< 0$ ; in-degree CI $\geq 0$      | $-0.08 \pm 0.03$ (19/27 $< 0$ ); 18/27            | partial; pass |
| P3 receptive $> 0$ ; specificity $< 0$      | $+0.00 \pm 0.02$ ; $-0.08 \pm 0.02$               | fail; pass    |
| P4 top-decile lift $\geq 2$ in $\geq 2/3$   | 9/27; median 1.78                                 | fail          |
| P5 read $>$ unread (in-flight)              | 7 testable; $-0.004 \pm 0.028$                    | inconclusive  |
| G26 leader deficit explained                | OR 0.22 raw, 0.28 [0.09, 0.53] adjusted           | failed        |
| G35 lead role adds nothing                  | OR 1.39 [0.86, 2.35]                              | supported     |
| NE42 transfer across merge, split           | $+33$ [+22, +44]; $-298$ millinats                | mixed         |



**Second figure.** (a) Synthetic validation on the real feature matrices of six periods. Under a kickoff bump and day drift, the pre-registered comparison (F vs B3) rejects a true null in up to 80% of runs; A1's baseline B4 holds size at 0.03. Power is 1.0 at  $\geq 2,400$  test ideas. (b) Pooled standardised coefficients of the seed features (random effects over 27 periods). Text gives the number of periods with CI below and above zero.

## GOAL PERIODS IN DETAIL

G06  $\times$  failed  
 G08  $\checkmark$  supported  
 G11  $\sim$  mixed  
 G12  $\times$  failed  
 G13  $\times$  failed  
 G16  $\times$  failed  
 G18  $\times$  failed  
 G19  $\times$  failed  
 G20  $\times$  failed  
 G21  $\sim$  mixed

+23 more periods: see the card.

## FAITHFULNESS SCORECARD

|                        |   |                   |   |
|------------------------|---|-------------------|---|
| A mapping              | 1 | F identifiability | 2 |
| B assumptions          | 1 | G ground truth    | 1 |
| C adequacy             | 1 | H comparative     | 1 |
| D unfitted predictions | 1 | I transfer        | 0 |
| E interventional       | 1 |                   |   |

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writup/paper.tex`).

## CAVEATS

Ideas are hashed markers, so long reports and code dumps coin many low-value names; length and novel load partly measure that. Reach counts co-generated uses. The in-flight placebo lacks power, because unread adopters are about 1% of adopters. Coefficients are in-sample; only  $\Delta LL$ , AUC and lift are held out. The 27 replication folders are one estimator, not 27 tests. A1 and A2 were written after the synthetic runs and before any real outcome was modelled. Some working timestamps were first written ahead of the real clock and were corrected from file times.

## ROUND 2

Run `confirm.py` on the #51 tail (dilution law and F-B4 gain), #15 and #28 after disclosure. Test agent-level poster fitness across periods. Locate each item inside its message (position vs length). Pool unread adopters hierarchically to test contagiousness against co-generation.

- **H61-R1** Fitness with per-agent pooling across periods (exception (b): check that poster effects are invariant first).
- **H61-R2** Within-message position: is an item spread less because the message is long, or because it sits late in it?
- **H61-R3** A powered convergence test: pool unread-5 adopters across periods with a hierarchical model, or use a longer in-flight window, so that co-generation can be separated from transmission.
- **H61-R4** The leader anomaly (G26): what in the leader's messages lowers uptake beyond length and load?